

initial guess at the joint angles θ^0 , and the tolerances ϵ_ω and ϵ_v on the final error. If a solution is not found within a maximum number of iterations, then `success` is false.

```
[thetalist,success] = IKinSpace(Slist,M,T,thetalist0,eomg,ev)
```

This is similar to `IKinBody`, except that the joint screws $\mathcal{S}_i$ are expressed in the space frame and the tolerances are interpreted in the space frame, also.

6.7 Notes and References

The inverse kinematics of the most general 6R open chain is known to have up to 16 solutions; this result was proved by Lee and Liang [87] and by Raghavan and Roth [143]. Procedures for finding closed-form inverse kinematics solutions to somewhat more general six-dof open chains than those treated in this chapter are described in [129, 122]; these procedures use solutions to a collection of some basic screw-theoretic subproblems, called the Paden–Kahan subproblems, e.g., finding the angle of rotation for a zero-pitch screw motion between a pair of given points. Iterative numerical procedures for finding all 16 solutions of a general 6R open chain are reported in [104].

A comprehensive summary of inverse kinematics methods for kinematically redundant robot arms are discussed in [26]. Many of these methods rely on results and solution techniques from least-squares optimization, and for this reason we provide a brief review of the basics of optimization in Appendix D; a classic reference for optimization is [98]. The repeatability (or cyclicity) conditions for a general class of inverse kinematic redundancy resolution schemes are examined in [162].

6.8 Exercises

Exercise 6.1 Write a program that solves the analytical inverse kinematics for a planar 3R robot with link lengths $L_1 = 3$, $L_2 = 2$, and $L_3 = 1$, given the desired position (x, y) and orientation θ of a frame fixed to the tip of the robot. Each joint has no joint limits. Your program should find all the solutions (how many are there in the general case?), give the joint angles for each, and draw the robot in these configurations. Test the code for the case of $(x, y, \theta) = (4, 2, 0)$.

Exercise 6.2 Solve the inverse position kinematics (you do not need to solve the orientation kinematics) of the 6R open chain shown in Figure 6.9.

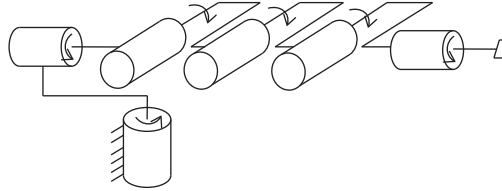


Figure 6.9: A 6R open chain.

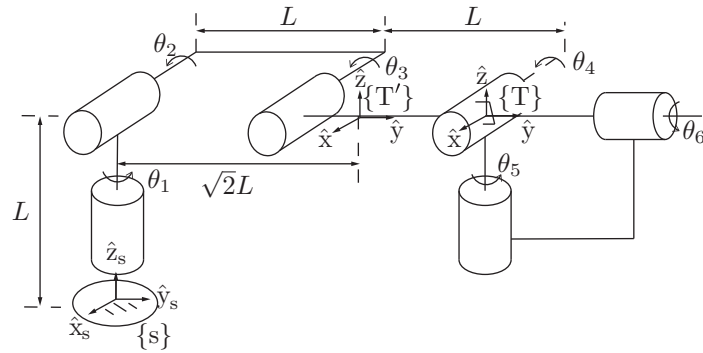


Figure 6.10: A 6R open chain.

Exercise 6.3 Find the inverse kinematics solutions when the end-effector frame $\{T\}$ of the 6R open chain shown in Figure 6.10 is set to $\{T'\}$ as shown. The orientation of $\{T\}$ at the zero position is the same as that of the fixed frame $\{s\}$, and $\{T'\}$ is the result of a pure translation of $\{T\}$ along the $\hat{y}_s$ -axis.

Exercise 6.4 The RRP open chain of Figure 6.11 is shown in its zero position. Joint axes 1 and 2 intersect at the fixed frame origin, and the end-effector frame origin p is located at $(0, 1, 0)$ when the robot is in its zero position.

- Suppose that $\theta_1 = 0$. Solve for θ_2 and θ_3 when the end-effector frame origin p is at $(-6, 5, \sqrt{3})$.
- If joint 1 is not fixed to zero but instead allowed to vary, find all the inverse kinematic solutions $(\theta_1, \theta_2, \theta_3)$ for the p given in (a).

Exercise 6.5 The four-dof robot of Figure 6.12 is shown in its zero position.

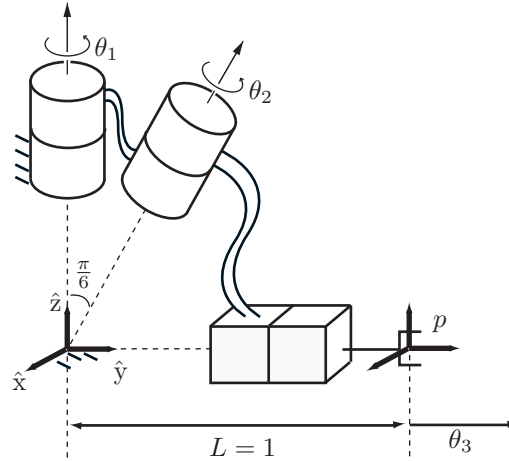


Figure 6.11: An RRP open chain.

Joint 1 is a screw joint of pitch h . Given the end-effector position $p = (p_x, p_y, p_z)$ and orientation $R = e^{[\hat{z}]\alpha}$, where $\hat{z} = (0, 0, 1)$ and $\alpha \in [0, 2\pi]$, find the inverse kinematics solution $(\theta_1, \theta_2, \theta_3, \theta_4)$ as a function of p and α .

Exercise 6.6 Figure 6.13(a) shows a surgical robot, which can be modeled as an RRP RP open chain as shown in Figure 6.13(b).

- In the general case, how many inverse kinematic solutions will exist for a given end-effector frame?
- Consider points A and B on the surgical robot shown in Figure 6.13(b). Given coordinates (x_A, y_A, z_A) and (x_B, y_B, z_B) for the points A and B in the fixed frame, find the joint variables $\theta_1, \theta_2, \theta_3, \theta_4$, and θ_5 . You should find an explicit formula for $(\theta_1, \theta_2, \theta_3)$ while for (θ_4, θ_5) you can just describe the procedure.

Exercise 6.7 In this exercise you are asked to draw a plot of a scalar $x_d - f(\theta)$ versus a scalar θ (similar to Figure 6.7) with two roots. Draw it so that, for some initial guess θ^0 , the iterative process actually jumps over the closest root and eventually converges to the further root. Hand-draw the plot and show the iteration process that results in convergence to the further root. Comment on the basins of attraction of the two roots in your plot.

Exercise 6.8 Use Newton–Raphson iterative numerical root finding to per-

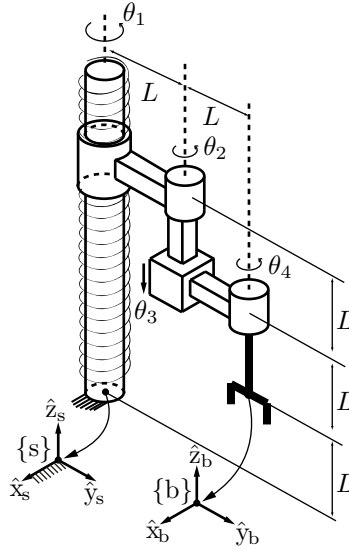


Figure 6.12: An open chain with a screw joint.

form two steps in finding the root of

$$g(x, y) = \begin{bmatrix} x^2 - 4 \\ y^2 - 9 \end{bmatrix}$$

when your initial guess is $(x^0, y^0) = (1, 1)$. Write the general form of the gradient (for any guess (x, y)) and compute the results of the first two iterations. You can do this by hand or write a program. Also, give all the correct roots, not just the one that would be found from your initial guess. How many are there?

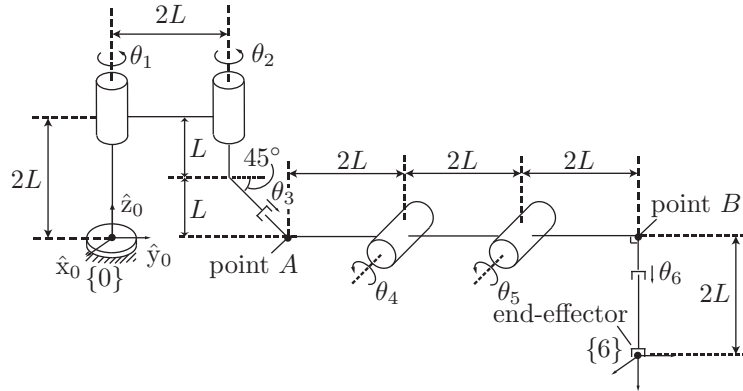
Exercise 6.9 Modify the function `IKinBody` to print out the results of each Newton–Raphson iteration, in a table similar to that for the 2R robot example in Section 6.2. Show the table produced when the initial guess for the 2R robot of Figure 6.8 is $(0, 30^\circ)$ and the goal configuration corresponds to $(90^\circ, 120^\circ)$.

Exercise 6.10 The 3R orthogonal axis wrist mechanism of Figure 6.14 is shown in its zero position, with joint axes 1 and 3 collinear.

- Given a desired wrist orientation $R \in SO(3)$, derive an iterative numerical procedure for solving its inverse kinematics.
- Perform a single iteration of Newton–Raphson root-finding using body-



(a) da Vinci S Surgical System instrument arm,
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(b) RRP RP robot at zero position.

Figure 6.13: Surgical robot and kinematic model.

frame numerical inverse kinematics. First write down the forward kinematics and Jacobian for general configurations of the wrist. Then apply your results for the specific case of an initial guess of $\theta_1 = \theta_3 = 0$, $\theta_2 = \pi/6$, with a desired end-effector frame at

$$R = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \in SO(3).$$

Exercise 6.11 The 3R nonorthogonal chain of Figure 6.15 is shown in its zero position.

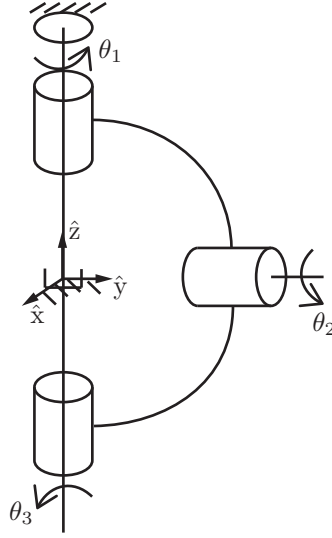


Figure 6.14: A 3R wrist.

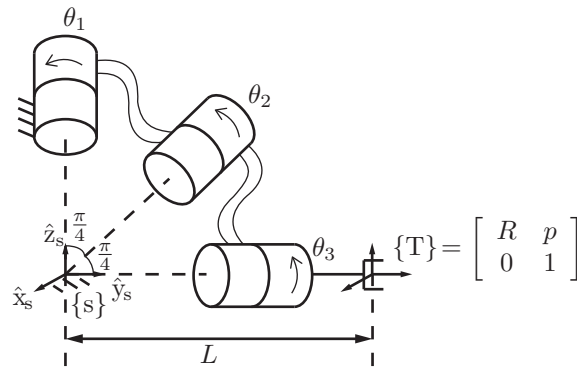


Figure 6.15: A 3R nonorthogonal chain.

- Derive a numerical procedure for solving the inverse position kinematics; that is, given some end-effector position p as indicated in the figure, find $(\theta_1, \theta_2, \theta_3)$.
- Given an end-effector orientation $R \in SO(3)$, find all inverse kinematic

solutions $(\theta_1, \theta_2, \theta_3)$.

Exercise 6.12 Use the function `IKinSpace` to find joint variables θ_d of the UR5 (Section 4.1.2) satisfying

$$T(\theta_d) = T_{sd} = \begin{bmatrix} 0 & 1 & 0 & -0.5 \\ 0 & 0 & -1 & 0.1 \\ -1 & 0 & 0 & 0.1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The distances are in meters. Use $\epsilon_\omega = 0.001$ rad (0.057°) and $\epsilon_v = 0.0001$ (0.1 mm). For your initial guess θ^0 , choose all joint angles as 0.1 rad. If T_{sd} is outside the workspace, or if you find that your initial guess is too far from a solution to converge, you may demonstrate `IKinBody` using another T_{sd} .

Note that numerical inverse kinematics is intended to find a solution close to the initial guess. Since your initial guess is not close to a solution (and remember that there are generally multiple solutions), the procedure may thrash about before finding a solution far from the initial guess. This solution may not respect joint limits. You can post-process the solution so that all joint angles are in the range $[0, 2\pi)$.

Exercise 6.13 Use the function `IKinBody` to find joint variables θ_d of the WAM (Section 4.1.3) satisfying

$$T(\theta_d) = T_{sd} = \begin{bmatrix} 1 & 0 & 0 & 0.5 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0.4 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Distances are in meters. Use $\epsilon_\omega = 0.001$ rad (0.057°) and $\epsilon_v = 0.0001$ (0.1 mm). For your initial guess θ^0 , choose all joint angles as 0.1 rad. If T_{sd} is outside the workspace, or if you find that your initial guess is too far from a solution to converge, you may demonstrate `IKinBody` using another T_{sd} .

Note that numerical inverse kinematics is intended to find a solution close to the initial guess. Since your initial guess is not close to a solution (and remember that there are generally multiple solutions), the procedure may thrash about before finding a solution far away from the initial guess. This solution may not respect joint limits. You can post-process the solution so that all joint angles are in the range $[0, 2\pi)$.

Exercise 6.14 The fundamental theorem of linear algebra (FTLA) states that,

given a matrix $A \in \mathbb{R}^{m \times n}$,

$$\begin{aligned}\text{null}(A) &= \text{range}(A^T)^\perp, \\ \text{null}(A^T) &= \text{range}(A)^\perp,\end{aligned}$$

where $\text{null}(A)$ denotes the null space of A (i.e., the subspace of $\mathbb{R}^n$ of vectors x that satisfy $Ax = 0$), $\text{range}(A)$ denotes the range or column space of A (i.e., the subspace of $\mathbb{R}^m$ spanned by the columns of A), and $\text{range}(A)^\perp$ denotes the orthogonal complement to $\text{range}(A)$ (i.e., the set of all vectors in $\mathbb{R}^m$ that are orthogonal to every vector in $\text{range}(A)$).

In this problem you are asked to use the FTLA to prove the existence of Lagrange multipliers (see Appendix D) for the equality-constrained optimization problem. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$, assumed differentiable, be the objective function to be minimized. The vector x must satisfy the equality constraint $g(x) = 0$ for given differentiable $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$.

Suppose that x^* is a local minimum. Let $x(t)$ be any arbitrary curve on the surface parametrized implicitly by $g(x) = 0$ (implying that $g(x(t)) = 0$ for all t) such that $x(0) = x^*$. Further, assume that x^* is a regular point of the surface. Taking the time derivative of both sides of $g(x(t)) = 0$ at $t = 0$ then leads to

$$\frac{\partial g}{\partial x}(x^*)\dot{x}(0) = 0. \quad (6.8)$$

At the same time, because $x(0) = x^*$ is a local minimum it follows that $f(x(t))$ (viewed as an objective function in t) has a local minimum at $t = 0$, implying that

$$\left. \frac{d}{dt}f(x(t)) \right|_{t=0} = \frac{\partial f}{\partial x}(x^*)\dot{x}(0) = 0. \quad (6.9)$$

Since (6.8) and (6.9) must hold for all arbitrary curves $x(t)$ on the surface defined by $g(x) = 0$, use the FTLA to prove the existence of a Lagrange multiplier $\lambda^* \in \mathbb{R}^m$ such that the first-order necessary condition,

$$\nabla f(x^*) + \frac{\partial g}{\partial x}(x^*)^T \lambda^* = 0,$$

holds.

Exercise 6.15

(a) For matrices A , B , C , and D , if A^{-1} exists show that

$$\begin{bmatrix} A & D \\ C & B \end{bmatrix}^{-1} = \begin{bmatrix} A^{-1} + EG^{-1}F & -EG^{-1} \\ -G^{-1}F & G^{-1} \end{bmatrix},$$

where $G = B - CA^{-1}D$, $E = A^{-1}D$, and $F = CA^{-1}$.

- (b) Use the above result to solve for the first-order necessary conditions for the equality-constrained optimization problem

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} x^T Q x + c^T x$$

subject to $Hx = b$, where $Q \in \mathbb{R}^{n \times n}$ is symmetric and positive definite and $H \in \mathbb{R}^{m \times n}$ is some matrix of maximal rank m . See Appendix D.